KL UNIVERSITY • BACHUPALLY CAMPUS, HYDERABAD
Department of Computer Science & Engineering

Credit Card Fraud Detection

Existing credit card fraud detection systems often struggle to detect new and complex fraud patterns, especially with highly imbalanced transaction data.

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AGENDA

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SECTION 1

Introduction & Domain Background

What is the domain?
Domain: FinTech & Cybersecurity
Credit card fraud detection uses machine learning to identify suspicious and unauthorized transactions.
Why is it relevant today?
The growth of digital payments has increased the need for fast and reliable fraud detection to protect users and financial institutions.
Where does your project sit?
Credit Card Fraud Detection uses machine learning to analyse transaction patterns and classify transactions as legitimate or fraudulent, with a frontend dashboard for monitoring and analysis.



Panel tip

Our project combines Machine Learning + Python Backend + Interactive Frontend to detect suspicious credit card transactions and provide quick, data-driven fraud analysis.

SECTION 2

Problem Statement & Motivation

PROBLEM STATEMENT

Existing fraud detection systems may fail to identify new and complex fraud patterns because they mainly rely on traditional methods, which can result in missed fraudulent transactions.

1

Who is affected?

Credit card users and banks are affected because undetected fraud can cause financial loss and security risks

2

What is the cost of the problem?

Financial losses, stolen money, and loss of customer trust due to undetected fraudulent transactions.

3

Why solve it now?

Online transactions are increasing, so credit card fraud is also increasing. Early detection can reduce financial losses and protect customers.

SECTION 3

Literature Survey

S. No	Author & Year	Title / Technique Used	Dataset / Tools	Key Findings	Limitation / Gap
1	Dal Pozzolo et al. (2017)	Credit Card Fraud Detection – Machine Learning	Real-world credit card transactions	ML can effectively detect fraudulent transactions and handle highly imbalanced data.	Fraud patterns change over time, making continuous adaptation difficult.
2	Carcillo et al. (2018)	SCARFF – Real-Time Fraud Detection using ML, Spark	Real-world streaming transactions; Kafka, Spark, Cassandra	Supports near real-time fraud detection and handles large transaction streams and class imbalance.	Requires complex Big Data infrastructure, which may be unnecessary for smaller systems.

S. No	Author & Year	Title / Technique Used	Dataset / Tools	Key Findings	Limitation / Gap
3	Fiore et al. (2019)	GAN-based Fraud Detection	Credit card transaction dataset; GAN + classifier	Generated minority-class samples improved fraud detection, especially sensitivity.	GAN-based methods are more complex and computationally demanding
4	Raghavan & El Bialy (2020)	Deep Learning Methods for Credit Card Fraud Detection	3 financial datasets; ML & Deep Learning	Deep learning methods showed strong performance compared with traditional ML methods.	Deep learning can require more computational resources and complex tuning..

S. No	Author & Year	Title / Technique Used	Dataset / Tools	Key Findings	Limitation / Gap
5	Van Houdt et al. (2020)	Autoencoder-based Fraud Detection	Benchmark credit card dataset	utoencoder reduced transaction features and improved fraud classification performance.	The approach is more complex than basic ML models.

SECTION 4

Research Gap Identified

GAP 1

Existing fraud detection methods may struggle with highly imbalanced transaction data, where fraudulent transactions are very few compared to legitimate ones.

GAP 2

Many existing methods focus mainly on detection accuracy and do not provide an easy-to-use interface for monitoring fraud.

GAP 3

Few systems combine machine learning detection, fraud alerts, transaction history, and analytics in one simple platform.

OUR PROJECT ADDRESSES

Our Credit Card Fraud Detection system addresses these gaps by combining Machine Learning-based fraud detection with an interactive dashboard for real-time analysis, alerts, transaction history, and analytics.

SECTION 5

Objectives of the Project

1. **To collect / prepare :** To collect and prepare a credit card transaction dataset containing legitimate and fraudulent transactions.

2. **To design & implement :** To design and implement a Machine Learning model for detecting fraudulent transactions.

3. **To evaluate :** To evaluate the model based on accuracy, precision, recall, and F1-score.

4. **To deploy / demonstrate :** To develop and demonstrate an interactive web application for fraud detection, alerts, and transaction analysis.

SECTION 6

Proposed System Architecture

INPUT

- **Transaction Details**
  - Time
  - Transaction amount
  - Transaction features (V1–V28)
- **Credit Card Dataset**
  - Legitimate transactions
  - Fraudulent transactions

User transaction data

PRE-PROCESSING

- Clean and prepare transaction data
- Handle missing/duplicate values
- Scale numerical features
- Handle class imbalance
- Split data into training and testing sets

Processed features

CORE MODULE

- Fraud Detection ML Model
- Feature analysis
- Model training
- Fraud classification
- Fraud probability / risk score
- Fraud alert generation

Predictions & results

OUTPUT

- Fraudulent / Legitimate result
- Fraud probability
- Risk level
- Transaction history
- Fraud alerts
- Analytics dashboard

1. **Transaction Data** (Time, amount, V1–V28 features, legitimate & fraud)
2. **Data Pre-processing** (Clean, handle missing, scale, balance, split train/test)
3. **Feature Selection & Scaling** (Choose important features, normalize)
4. **ML Model Training** (Train algorithms, (optimize parameters)
5. **Fraud Detection** (Classify transaction as fraud / legitimate)
6. **Risk Score / Classification** (Probability & risk level)
7. **Dashboard & Fraud Alerts** (History, analytics, real-time alerts)

SECTION 7

Methodology & Modules

1 Module 1 - Data

Credit card transaction dataset containing transaction time, amount, transaction features (V1–V28), and fraud/legitimate labels. The dataset is divided into training and testing data.

2 Module 2 - Processing

Data cleaning**,** handling missing values, feature scaling, class balancing, and preparation of data for the ML model.

3 Module 3 - Core Logic

Machine Learning model analyses transaction patterns and classifies transactions as Fraudulent or Legitimate, along with fraud probability/risk score.

4 Module 4 - Interface

Frontend: HTML, CSS, JavaScript
Backend: Python Flask
The user enters transaction details and views fraud results, alerts, transaction history, and analytics through the web dashboard.

SECTION 8

Tools and Technologies

SOFTWARE

Language: Python 3.11
Libraries: Pandas, NumPy, Scikit-learn, Joblib
Front End: HTML, CSS, JavaScript
Back End: Python Flask
Database: SQLite
IDE: VS Code
Visualization: Chart.js / Matplotlib

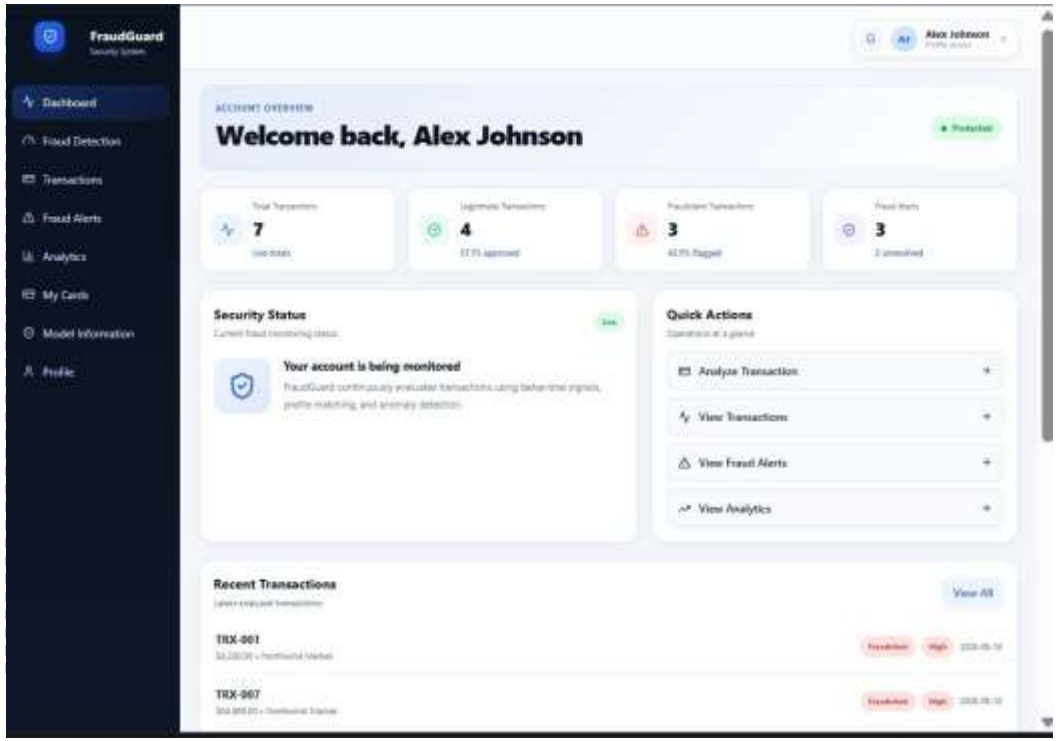
HARDWARE / PLATFORM

Processor: Intel Core i5 or above
RAM: 8 GB minimum
GPU / Cloud: Not required / Google Colab if needed
Deployment: Localhost
Version Control: GitHub
Platform: Windows 11

SECTION 9

Work Completed / Demo

Literature survey	Completed
Dataset collection	Completed
Pre-processing pipeline	Completed
Model training	Completed
User interface	In progress
Testing & evaluation	In progress



SECTION 10

Timeline & Review Plan

Review 0

Week 2

- Problem understanding and project scope
- Dataset selection plan
- Team roles finalized
- Initial proposal, abstract and objectives

Review 1

Week 5

- Literature survey completed
- Dataset collection and exploratory analysis
- Pre-processing pipeline developed

Review 2

Week 9

- System architecture finalized
- Model training and tuning
- Stream lit interface development

Final

Week 14

- Complete system with all features
  Final testing and evaluation
  Full demo

SECTION 11

Expected Outcome & Conclusion

DELIVERABLE

A working web-based Credit Card Fraud Detection System that detects and classifies transactions as fraudulent or legitimate.

MEASURABLE TARGET

Achieve high fraud-detection performance based on accuracy, precision, recall, and F1-score, with fast prediction response.

BEYOND THE LAB

Can be used by banks and financial institutions to monitor transactions and identify suspicious activities.

CONCLUSION

Our project aims to detect fraudulent credit card transactions using Machine Learning. Existing methods face challenges such as imbalanced data and changing fraud patterns. Our system provides fraud detection, risk analysis, alerts, transaction history, and an interactive dashboard.
Future scope: Real-time transaction monitoring and advanced deep-learning models can be add

SECTION 12

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Thank You

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